

Task Name:- AIML Assignment- 13

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Domain:- AI/ML

Branch:- Computer Engineering

Collage:- Dr. Panjabrao Deshmukh Polytechnic, Amravati.

Question no: 1

Program:

```
import numpy as np

# 1d Array

arr1 = np.array([10, 20, 30, 40, 50])


#2d Array

arr2 = np.array([[1, 2, 3],
                 [4, 5, 6],
                 [7, 8, 9]])


print("1D Array:")
print(arr1)
print("Shape:", arr1.shape)
print("Data Type:", arr1.dtype)


print("1D Array:")
print(arr1)
print("Shape:", arr1.shape)
print("Data Type:", arr1.dtype)
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question1.py"
1D Array:
[10 20 30 40 50]
Shape: (5,)
Data Type: int64
1D Array:
[10 20 30 40 50]
Shape: (5,)
Data Type: int64
PS E:\ITR 2026 AIML\day13 task> █
```

Question 2:

Program:

```
import numpy as np
```

```
zeros1 = np.zeros(8)
```

```
ones = np.ones((4, 4))
```

```
zeros2 = np.zeros((3, 3))
```

```
print("1D Array of Zeros")
```

```
print(zeros1)
```

```
print("\n4x4 Ones Matrix")
```

```
print(ones)
```

```
print("\n3x3 Zero Matrix")
```

```
print(zeros2)
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question2.py"
1D Array of Zeros
[0. 0. 0. 0. 0. 0. 0. 0.]

4x4 Ones Matrix
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]]

3x3 Zero Matrix
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
PS E:\ITR 2026 AIML\day13 task>
```

Question no3:

Program:

```
import numpy as np
```

```
a = np.arange(0, 21)
```

```
b = np.arange(10, 51, 2)
```

```
c = np.arange(5, 101, 5)
```

```
print("0 to 20")
```

```
print(a)
```

```
print("\nEven Numbers")
```

```
print(b)
```

```
print("\nMultiples of 5")
```

```
print(c)
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question3.py"
0 to 20
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]

Even Numbers
[10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50]

Multiples of 5
[  5  10  15  20  25  30  35  40  45  50  55  60  65  70  75  80  85  90
 95 100]
PS E:\ITR 2026 AIML\day13 task>
```

Question no 4:

Program:

```
import numpy as np
```

```
arr1 = np.linspace(0, 5, 10)
```

```
arr2 = np.linspace(-10, 10, 15)
```

```
print(arr1)
```

```
print("Length:", len(arr1))
```

```
print()
```

```
print(arr2)
```

```
print("Length:", len(arr2))
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question4.py"
[0.          0.55555556 1.11111111 1.66666667 2.22222222 2.77777778
 3.33333333 3.88888889 4.44444444 5.          ]
Length: 10

[-10.         -8.57142857 -7.14285714 -5.71428571 -4.28571429
 -2.85714286 -1.42857143  0.          1.42857143  2.85714286
  4.28571429  5.71428571  7.14285714  8.57142857 10.          ]
Length: 15
PS E:\ITR 2026 AIML\day13 task> 
```

Question no5:

Program:

```
import numpy as np

a = np.random.rand(10)

b = np.random.randn(3, 3)

c = np.random.randint(10, 51, (4, 5))

print("Random Numbers")

print(a)

print("\nStandard Normal Matrix")

print(b)

print("\nRandom Integers")

print(c)
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question5.py"
Random Numbers
[0.89600737 0.24450181 0.21631898 0.25348484 0.34298398 0.27956374
 0.52319784 0.20924821 0.69992679 0.07579208]

Standard Normal Matrix
[[-0.09696658  0.32186717  0.9981968 ]
 [ 0.57636754  0.35738519  1.0435022 ]
 [-1.06841288  0.65565649  1.92730241]]

Random Integers
[[48 32 16 33 24]
 [36 38 50 21 50]
 [31 30 22 42 35]
 [37 22 31 20 36]]
PS E:\ITR 2026 AIML\day13 task>
```

Question no 6:

Program:

```
import numpy as np

v1 = np.array([2, 4, 6, 8])

v2 = np.array([1, 3, 5, 7])

print("Addition")
print(v1 + v2)

print("\nSubtraction")
print(v1 - v2)

print("\nElement-wise Multiplication")
print(v1 * v2)

print("\nDot Product")
print(np.dot(v1, v2))
```

Output

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question6.py"
Addition
[ 3  7 11 15]

Subtraction
[1 1 1 1]

Element-wise Multiplication
[ 2 12 30 56]

Dot Product
100
PS E:\ITR 2026 AIML\day13 task> █
```

Question no 7

Program:

```
import numpy as np
```

```
A = np.array([[1,2,3],  
              [4,5,6],  
              [7,8,9]])
```

```
B = np.array([[9,8,7],  
              [6,5,4],  
              [3,2,1]])
```

```
print("Addition")
```

```
print(A+B)
```

```
print("\nMatrix Multiplication")
```

```
print(A@B)
```

```
print("\nElement-wise Multiplication")
```

```
print(A*B)
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question7.py"  
Addition  
[[10 10 10]  
 [10 10 10]  
 [10 10 10]]  
  
Matrix Multiplication  
[[ 30  24  18]  
 [ 84  69  54]  
 [138 114  90]]  
  
Element-wise Multiplication  
[[ 9 16 21]  
 [24 25 24]  
 [21 16  9]]  
PS E:\ITR 2026 AIML\day13 task> █
```

Question no 8

Program:

```
import numpy as np

arr = np.random.randint(1,101,(4,4))

print(arr)

print("Shape:", arr.shape)

print("Dimension:", arr.ndim)

print("Size:", arr.size)

print("Data Type:", arr.dtype)

print("Minimum:", arr.min())

print("Maximum:", arr.max())
```

Output:

```
PS E:\ITR 2026 AIMA\day13 task> python -u "e:\ITR 2026 AIMA\day13 task\Question8.py"
[[66 89 63 50]
 [29  7 26 98]
 [97 73 40 58]
 [55 66 32 45]]
Shape: (4, 4)
Dimension: 2
Size: 16
Data Type: int32
Minimum: 7
Maximum: 98
PS E:\ITR 2026 AIMA\day13 task>
```


Question 9

Program:

```
import numpy as np

arr = np.random.randint(1,51,20)

matrix = arr.reshape(4,5)

print(matrix)

print("Sum:", matrix.sum())

print("Mean:", matrix.mean())

print("Standard Deviation:", matrix.std())

print("Maximum in each row")

print(matrix.max(axis=1))
```

Output:

```
PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question9.py"
[[40  1 14 22 49]
 [37 47  3 18 14]
 [19 10 19 49 21]
 [46 21 37 36  3]]
Sum: 506
Mean: 25.3
Standard Deviation: 15.624019969265271
Maximum in each row
[49 47 49 46]
PS E:\ITR 2026 AIML\day13 task> █
```

Question no 10

Program:

```
import numpy as np

try:
    n = int(input("How many numbers do you want to generate? "))

    arr = np.random.randint(10,101,n)

    print("\nGenerated Array")
    print(arr)

    print("\nMean:", np.mean(arr))

    print("Median:", np.median(arr))

    print("Standard Deviation:", np.std(arr))

    print("Minimum:", np.min(arr))

    print("Maximum:", np.max(arr))

    if n % 2 == 0:
        matrix = arr.reshape(2, n//2)

        print("\nReshaped Array")

        print(matrix)
```

```

print("\nRow-wise Sum")

print(np.sum(matrix, axis=1))

else:

    print("\nCannot reshape into a 2D array because the number of elements is odd.")

except ValueError:

    print("Please enter a valid integer.")

except Exception as e:

    print("Error:", e)

```

Outputs:

```

PS E:\ITR 2026 AIML\day13 task> python -u "e:\ITR 2026 AIML\day13 task\Question10.py"
How many numbers do you want to generate? 4

Generated Array
[99 42 56 10]

Mean: 51.75
Median: 49.0
Standard Deviation: 31.971667144520318
Minimum: 10
Maximum: 99

Reshaped Array
[[99 42]
 [56 10]]

Row-wise Sum
[141 66]
PS E:\ITR 2026 AIML\day13 task> 

```